

New Test Case: The Smart Home Sensor Conflict

This scenario tests the engine's ability to resolve conflicts in a chain of reasoning when new evidence contradicts a deeply nested conclusion.

1. Initial Belief Base (KB)

We define a belief base representing a smart home security system. Each formula is assigned a priority (1 = Low, 10 = High).

- **Rule 1 (R_1):** $S \rightarrow A$ (If the sensor is tripped, the alarm sounds) — **Priority: 9**
- **Rule 2 (R_2):** $W \rightarrow S$ (If a window is opened, the sensor trips) — **Priority: 8**
- **Fact 1 (F_1):** W (The window is currently open) — **Priority: 2**

Current State:

Your code should currently entail S (Sensor tripped) and A (Alarm sounding) via the chain $W \rightarrow S \rightarrow A$.

2. Operation: Revision with $\neg A$

The user provides new information: "The alarm is not sounding" ($\neg A$). To maintain consistency, the engine must perform a revision: $B * \neg A$.

Step 1: Contraction ($B \div A$)

To accept $\neg A$, the engine must first ensure A is no longer entailed. To break the logical chain leading to A , the engine has three options for removal:

1. Remove R_1 ($S \rightarrow A$) — Cost: 9
2. Remove R_2 ($W \rightarrow S$) — Cost: 8
3. Remove F_1 (W) — Cost: 2

Expected Behavior of your code:

Because F_1 (Fact: Window is open) has the lowest priority, your code should select it for removal to satisfy the *Inclusion-maximal* remainder requirement while minimizing "epistemic loss."

Step 2: Expansion

The engine adds the new information $\neg A$ to the remaining base.

3. Expected Final Belief Base

The resulting Belief Base should be:

$$\{S \rightarrow A, W \rightarrow S, \neg A\}$$

4. Verification of Post-Conditions

To confirm the engine has revised correctly, the following checks should be performed:

- **Success Postulate:** Is $\neg A$ entailed? (**Yes**)
- **Consistency:** Is the base consistent? (**Yes**)
- **New Entailment (Modus Tollens):** Due to the remaining rules, your code should now be able to entail $\neg S$ (The sensor is not tripped) and $\neg W$ (The window is not open).